Like Parent, Like Child? The Intergenerational Transmission of Hand-to-Mouth Status

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Abstract

Why do some households remain persistently close to the liquidity constraint while others accumulate financial buffers? Using genealogical links in the PSID, I document substantial intergenerational persistence in hand-to-mouth (HtM) status, with a parent–child correlation of 0.45 in lifetime HtM rates. This persistence reflects inherited wealth, persistent earnings capacity, and a component that family resources do not explain. I develop a dynastic heterogeneous-agent incomplete-markets model incorporating three corresponding channels of intergenerational transmission: inherited wealth, earnings capacity, and patience. Estimating the model by the simulated method of moments, I find that inherited wealth accounts for 61 percent of intergenerational HtM persistence, while inherited patience and earnings capacity account for 21 and 18 percent, respectively. These dynastic mechanisms also shape macroeconomic policy outcomes. Increasing unemployment-insurance generosity produces a larger steady-state welfare gain in the estimated economy than in a one-generation incomplete-markets benchmark. Finally, HtM households exhibit substantially larger consumption responses to aggregate productivity shocks.

Keywords: Hand-to-Mouth; Intergenerational Mobility; Liquidity; Heterogeneous Agents;

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1 Introduction

Why do some households remain persistently hand-to-mouth (HtM), while others accumulate substantial financial buffers? This question is central to macroeconomics because HtM households play a disproportionate role in consumption dynamics, fiscal transmission, and the welfare effects of social insurance. Yet most of the existing literature studies HtM status within a single generation, leaving open whether liquidity constraints reflect temporary responses to shocks or economic circumstances that persist across generations.

Using genealogical links from the Panel Study of Income Dynamics (PSID), I document substantial intergenerational persistence in HtM status in the United States. Children whose parents spend more of their adult lives HtM are substantially more likely to be HtM themselves, with a parent–child correlation in lifetime HtM rates of 0.45. I then examine what lies behind this correlation, and the data point to at least three sources. The first is inherited wealth. Children of parents in the bottom wealth quintile spend roughly 74 percent of their adult lives HtM, compared with 32 percent for children of parents in the top quintile. The second is persistent earnings capacity. Income is strongly persistent across generations, with a rank–rank slope of 0.41, and children of parents in the bottom income quintile spend 68 percent of their adult lives HtM, compared with 30 percent at the top. The third component cannot be explained by observable family resources. However, the parent–child relationship in HtM status remains large and statistically significant after controlling for parental wealth, income, and demographics.

To account for these three patterns, I develop a dynastic heterogeneous-agent incomplete-markets model in the spirit of Aiyagari (1994) with three corresponding channels of intergenerational transmission. To capture inherited wealth, accidental bequests transmit assets across generations, so children begin adulthood with part of their parents balance sheet. To capture persistent earnings capacity, labor productivity includes a permanent component that children inherit from their parents. Separating this component from transitory fluctuations in the PSID, I estimate a parent–child correlation of 0.56. For the persistence that family resources do not explain, I introduce heterogeneous discount factors that are transmitted across generations. Patience is a natural candidate for this residual. It is strongly related to wealth accumulation (Epper et al., 2020), and time preferences are correlated between parents and children (Brenoe and Epper, 2022), so children of impatient parents can remain close to the liquidity constraint even when they inherit similar resources.

I estimate the model by the simulated method of moments, targeting the bequest-to-income ratio, the parent–child correlation in HtM rates, the wealth Gini, and the mean wealth-to-income ratio. The three mechanisms jointly account for the observed HtM persistence, while a restricted dynastic model with a single discount factor and no inherited

earnings capacity generates only about half of the observed parent–child HtM correlation.

I use the estimated model to quantify the importance of each channel by shutting down the mechanisms individually and in combination, re-solving general equilibrium in each counterfactual. A Shapley decomposition attributes 61 percent of intergenerational HtM persistence to inherited wealth, 21 percent to inherited patience, and 18 percent to inherited earnings capacity. Removing all three channels eliminates intergenerational HtM persistence in the model.

These dynastic mechanisms also matter for macroeconomic policy. Increasing the unemployment-insurance replacement rate from 0.30 to 0.35 raises stationary-equilibrium welfare by 0.45 percent of consumption in the estimated economy, compared with 0.04 percent in a one-generation incomplete-markets benchmark calibrated to have a similar initial equilibrium interest rate and capital–labor ratio. When both policy regimes are evaluated using each economy's initial distribution of household states, the welfare gains are 0.51 and 0.38 percent, respectively. The difference between the stationary-equilibrium results largely reflects the endogenous adjustment of the distribution of household states, which offsets more of the reform's benefits in the one-generation economy.

I also examine consumption responses to an unexpected aggregate productivity shock. Following a one-percent increase in aggregate productivity, HtM households raise consumption by 1.16 percent on impact, nearly twice as much as non-HtM households, while impatient households respond about 56 percent more than patient households. These results illustrate how inherited differences in resources and saving behavior shape the distribution of consumption responses to aggregate fluctuations.

This paper connects the literature on HtM households and consumption responses to liquidity constraints (Zeldes, 1989; Kaplan and Violante, 2014; Kaplan et al., 2014; Aguiar et al., 2025; Auclert, 2019) with the literature on the intergenerational transmission of economic status (Becker and Tomes, 1979, 1986; Solon, 1992; Charles and Hurst, 2003; Chetty et al., 2014; Fagereng et al., 2021). While the former emphasizes how the distribution of constrained households shapes consumption and policy transmission, the latter documents substantial persistence in income and wealth across generations. Recent work also shows that earnings dynamics and risk are transmitted across generations (Halvorsen et al., 2022). I bring these perspectives together by asking whether a household's position relative to the liquidity constraint is itself persistent across generations and by quantifying the roles of inherited wealth, earnings capacity, and patience. The paper is also related to quantitative models of dynastic linkages (De Nardi, 2004; Boar, 2021; Lee and Seshadri, 2019) and to work studying unemployment insurance and consumption smoothing in incomplete-markets economies (Krusell et al., 2010; Kekre, 2023; Ganong and Noel, 2019). The paper's contribution is to show that where households sit relative to the liquidity constraint is partly

inherited, and that this matters for policy.

The rest of the paper proceeds as follows. Section 2 describes the PSID data. Section 3 constructs the HtM measures and empirically documents the persistence of HtM. Section 4 presents the dynastic model. Section 5 discusses calibration and estimation. Section 6 presents the decomposition and policy counterfactuals. Section 7 examines the sensitivity of the results. Section 8 concludes.

2 Data: PSID Sample and Parent-Child Links

The empirical analysis uses the Panel Study of Income Dynamics (PSID) from 1984 to 2023. The PSID is well suited to studying intergenerational HtM persistence because its genealogical structure allows individuals who form their own households to remain linked to their parents. I use these links to construct parent–child pairs and combine the asset, income, and employment histories of both generations.

I measure HtM exposure over working-age observations, restricting both parents and children to ages 30 to 65 who are in the PSID for at least 3 waves. For each individual, I construct a lifetime HtM rate equal to the share of observed working-age survey waves in which the household is classified as HtM. The resulting dataset is organized at the child level and links each child's HtM history to the corresponding histories and characteristics of the parent generation. The income-process estimation uses the narrower age range of 30 to 55.

In addition, I drop observations with annual real income or expenditure below $2,000, all nominal variables are converted to real 2009 dollars using the CPI-U, and all statistics are weighted using PSID family weights.

2.1 Variable Construction

The main variables are household income, liquid wealth, total net wealth, and intergenerational transfers. Liquid wealth consists of balances in checking and savings accounts and certificates of deposit. Total net wealth includes both liquid and illiquid assets, including home equity, and illiquid wealth is defined as total net wealth less liquid wealth. Household income is measured using annual labor income of the household head.

I measure intergenerational transfers using observed inheritances and inter vivos transfers received by each child. Because transfers are received at different ages while the model represents inheritance as a single transfer when born, I express observed transfers as a common present value. Appendix A.1 provides additional details on the data construction and describes the transfer construction.

Using these variables, I construct indicators for HtM status and corresponding lifetime rates. A household's HtM rate is the fraction of its observed working-age years spent close to the liquidity constraint and is the main measure of financial vulnerability used in the intergenerational analysis. [1]

3 Empirical Evidence

This section shows the reduced-form relationships that motivate the three channels of the dynastic model. I first show that HtM rates are persistent across generations and then present evidence on wealth concentration, income persistence, and saving heterogeneity.

3.1 Hand to Mouth Status in the PSID

I begin by defining HtM status in the data. The key idea is that a household is HtM if it has limited liquid wealth relative to its current income, and therefore little ability to smooth consumption in response to income fluctuations.

I classify a household as HtM if liquid wealth is less than -0.165 times annual income, or if liquid wealth is nonnegative but no greater than $\frac{1}{26}$ of annual income. Among these households, I define a household as *poor HtM* if its illiquid wealth is at most $\frac{1}{6}$ of annual income, equivalent to two months of income, and as *wealthy HtM* if its illiquid wealth exceeds that threshold. Under this definition 46.9 percent of working-age household-years are HtM, split into 20.6 percent poor HtM and 26.3 percent wealthy HtM.

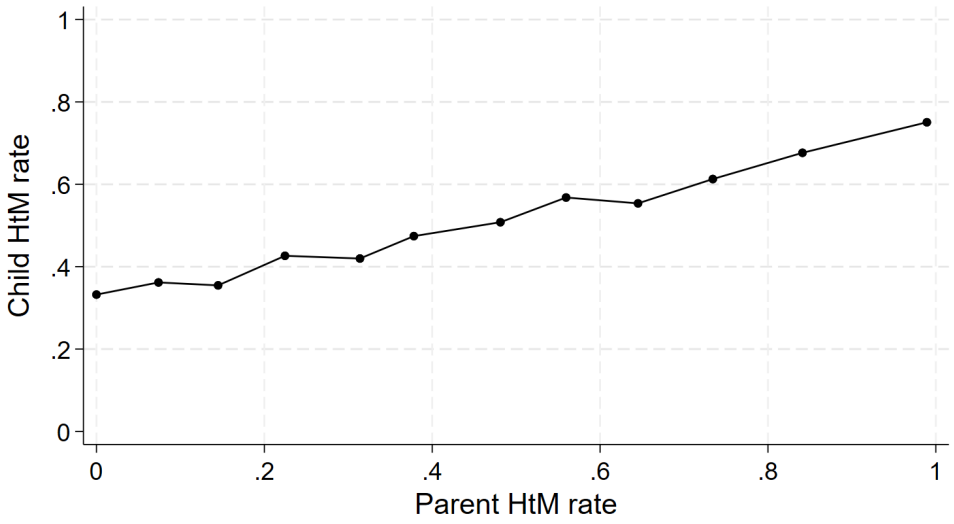
Appendix A.2 compares this definition with alternatives in the literature and reports the correlations among them, reports summary statistics for each group, and shows that HtM status is persistent within generations.

3.2 Intergenerational Persistence

The central empirical fact of the paper is that HtM exposure is persistent across generations. Figure 1 shows a strong positive relationship between parent and child lifetime HtM rates. Children whose parents spend a larger share of their adult lives HtM tend to do so themselves. The parent–child correlation is 0.447 (s.e. 0.014), providing the key intergenerational persistence moment targeted in the quantitative model.

[1] The estimation targets the union of poor and wealthy HtM status. The model has a single liquid asset, so it generates households with low liquid balances but cannot distinguish those who also lack illiquid wealth from those who do not.

Figure 1. Binned relationship between parent and child HtM rates over adult life



To examine how much of this persistence is associated with observable family characteristics, Table 1 regresses the child's lifetime HtM rate on the parent's, progressively controlling for parental liquid wealth, income, and demographics. Controlling for parental liquid wealth alone reduces the coefficient by roughly half, from 0.447 to 0.227. Part of this reduction reflects the close relationship between the two variables, since the parent's HtM rate is itself defined using liquid wealth relative to income. Nevertheless, substantial persistence remains after controlling for wealth. The coefficient is 0.214 after adding parental income and 0.179 after adding demographics, remaining statistically significant in every specification. Although these regressions cannot disentangle the causal effects of the different transmission channels, they highlight the importance of family resources and show that intergenerational HtM persistence remains substantial even after controlling for observable characteristics.

Table 1. Intergenerational Persistence in HtM Status, with Controls

	Dependent variable: child HtM rate			
	(1) Raw	(2) +Wealth	(3) +Income	(4) +Demographics
Parent HtM rate	0.447***	0.227***	0.214***	0.179***
	(0.022)	(0.037)	(0.036)	(0.034)
IHS parent liquid wealth		−0.035***	−0.029***	−0.014***
		(0.004)	(0.004)	(0.004)
Log parent income			−0.035***	0.006
			(0.008)	(0.008)
Observations	3,390	3,390	3,390	3,390
Family clusters	1,361	1,361	1,361	1,361
R^2	0.207	0.234	0.243	0.346

Notes: One observation per adult child. Standard errors, clustered on family identifier. Estimates are weighted by PSID family weights. The sample requires both the child and at least one parent to be observed for more than three waves. Demographics include education, age, race, and sex. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

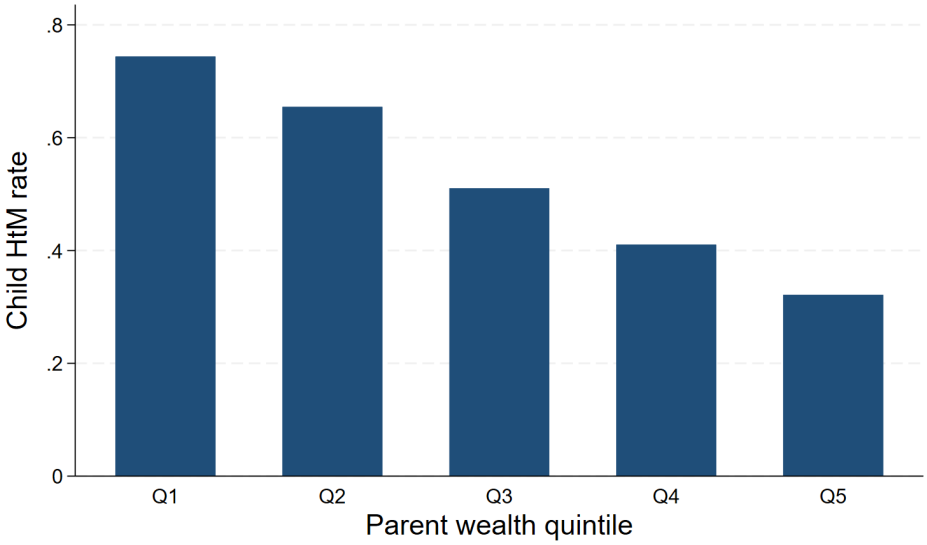
3.3 Evidence on the Three Transmission Channels

Having established that HtM rates are persistent across generations, I turn to the mechanisms that may generate this persistence. Three empirical patterns motivate the channels in the model: the transmission of wealth, the persistence of earnings capacity, and persistent heterogeneity in saving behavior.

First, wealth is highly concentrated and transmitted across generations. The average wealth Gini in the PSID between 1984–2007 is 0.728, and among recipients the present value of inheritances and inter vivos transfers averages 2.54 times the heir's income at the start of working life. Consistent with wealth providing an important buffer against liquidity constraints, Figure 2 shows a strong gradient in HtM exposure by parental wealth. Children of parents in the bottom wealth quintile spend roughly 74 percent of their adult lives HtM, compared with about 32 percent among children of parents in the top quintile.

Second, earnings capacity is persistent across generations and parental earnings predict children's HtM status. Measuring income as average household labor income over years with positive earnings, Figure 3(a) shows a close to linear relationship between parent and child income ranks, with a rank–rank slope of 0.41 (s.e. 0.02). The corresponding intergenerational elasticity of income is 0.34 (s.e. 0.03) Parental income is also strongly associated with children's liquidity. Figure 3(b) shows that children of parents in the bottom income quintile spend 68 percent of their adult lives HtM, compared with 30 percent among

Figure 2. Child HtM rates by parent wealth quintile

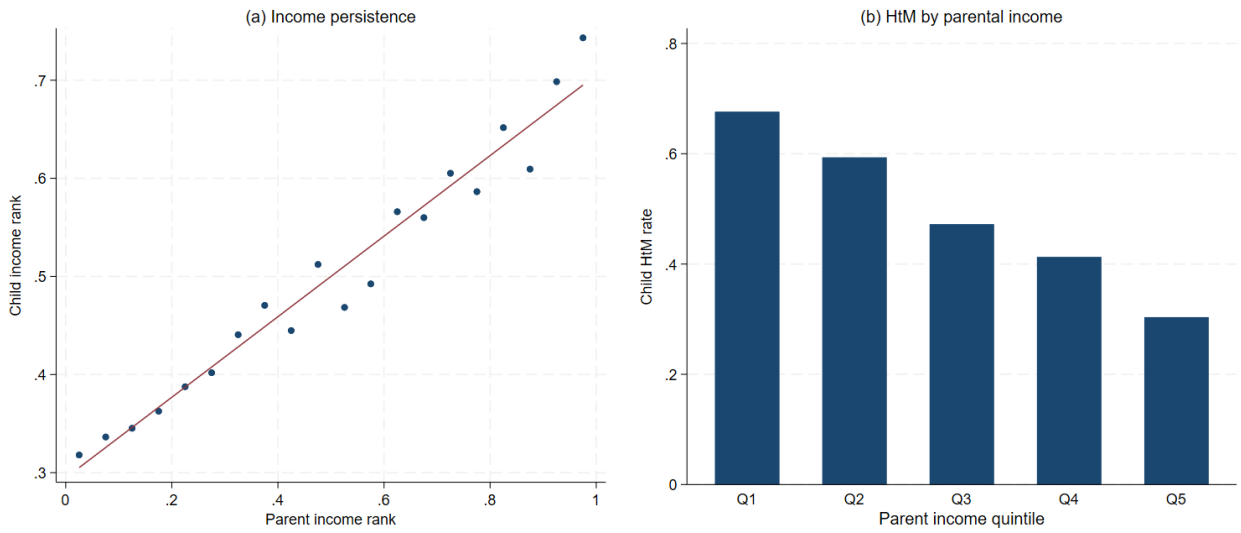


children of parents in the top quintile, a gradient nearly as steep as the one by parental wealth in Figure 2.

Third, persistence in HtM status is not fully accounted for by family resources. Table 1 shows that the parent–child HtM coefficient remains 0.179 and statistically significant after controlling for parental wealth, income, and demographics. A natural explanation is persistent heterogeneity in saving behavior. Patience is strongly related to wealth accumulation (Epper et al., 2020), and economic preferences are correlated between parents and children (Brenoe and Epper, 2022). I therefore allow discount-factor types to be transmitted across generations and estimate the strength of this channel structurally.

Together, these facts motivate the three dynastic channels in the model: wealth, earnings capacity, and patience.

Figure 3. Intergenerational persistence of income and HtM status by parental income



Notes: Panel (a) is a binned scatter plot of child income rank against parent income rank (20 bins). The fitted slope is 0.41 (s.e. 0.02). Panel (b) shows the average lifetime HtM rate of children by quintile of parental income. Income is average household labor income over years with positive earnings. Sample of 3,318 children observed for at least three waves with a parent observed for at least three waves, weighted by PSID family weights.

4 A Dynastic Model of Hand-to-Mouth Persistence

The benchmark model is a stationary, infinite-horizon general-equilibrium incomplete-markets model in the tradition of Aiyagari (1994), extended with dynastic transmission. Households face uninsurable idiosyncratic earnings and unemployment risk and can smooth consumption through a single risk-free asset.[2] A representative firm rents capital and hires labor in competitive markets, and a government runs a balanced-budget unemployment-insurance scheme. I extend the environment along three dimensions that let characteristics persist across generations: (i) intergenerational transfers of wealth, (ii) heterogeneous and inherited discount factors, and (iii) an inherited permanent component of labor productivity.

4.1 Households

The economy is populated by a continuum of dynasties of unit measure. A household is characterized by assets a, a labor-income state $s = (\xi, z, e)$ consisting of permanent productivity ξ, transitory productivity z, and employment status $e \in \{U, E\}$, and a discount-factor type $\beta \in \{\beta_L, \beta_H\}$. Discount-factor heterogeneity is represented by two types centered around $\bar{\beta}$,

[2] As emphasized by Kaplan and Violante (2022), targeting the level of liquid assets rather than total assets generates realistic marginal propensities to consume but then cannot match the top of the wealth distribution, consistent with a less tractable model with both illiquid and liquid assets.

$$\beta_L = \bar{\beta} - \sigma_\beta, \qquad \beta_H = \bar{\beta} + \sigma_\beta, \tag{1}$$

where σ_β governs the dispersion in patience.

Each period a household survives with probability $1 - p_d$ and dies with probability p_d. Conditional on surviving, the household chooses consumption c and next-period assets a' to solve

$$V(a, s; \beta) = \max_{a' \geq \bar{a}} \Big\{ u(c) + \beta\,(1 - p_d)\,\mathbb{E}\Big[V(a', s'; \beta) \,\Big|\, s\Big] \Big\}, \tag{2}$$

subject to the budget constraint

$$c + a' = R\,a + n(s), \qquad n(s) = \begin{cases} (1 - \tau_l)\,w\,y(s) & \text{if employed,} \\ b_u & \text{if unemployed,} \end{cases} \tag{3}$$

where R is the gross risk-free return, w the wage, $\bar{a} = 0$ the borrowing limit, τ_l a proportional labor-income tax that funds unemployment benefits, and b_u the benefit.

Death ends the household's problem. Equation (2) contains no continuation value from leaving wealth, so bequests are accidental. They arise from the assets a household happens to hold when it dies rather than from a deliberate bequest motive.

Upon death, a household is replaced by a single descendant, keeping the population stationary. The descendant receives an accidental bequest

$$a'_0 = (1 - \tau_b)\,a', \tag{4}$$

where $\tau_b \in [0, 1]$ is the share of the estate that does not reach the heir, capturing probate and transfer costs, end-of-life expenditure, estate taxation, and shares passing to relatives other than the tracked descendant. The descendant also draws a discount-factor type from $P_\beta(\cdot \mid \beta)$, inherits a permanent productivity component through the birth transition matrix Q_ξ of Section 4.5, draws a transitory component from its stationary distribution, and enters employed. Surviving households retain their discount-factor type.

4.2 Firms

A representative firm produces the final good using a Cobb–Douglas technology,

$$Y = Z\,K^\alpha L^{1-\alpha}, \tag{5}$$

where Z is total factor productivity, normalized to $Z = 1$ in the stationary equilibrium, K is aggregate capital, L aggregate effective labor, and $\alpha \in (0, 1)$ the capital share. Capital depreciates at rate δ_K.

The firm operates in competitive factor markets, renting capital and hiring labor to maximize profits,

$$\max_{K,L} \; Z\,K^{\alpha}L^{1-\alpha} - wL - (R - 1 + \delta_K)K. \tag{6}$$

The first-order conditions equate factor prices to marginal products,

$$R = \alpha Z \left(\frac{K}{L}\right)^{\alpha-1} + 1 - \delta_K, \qquad w = (1-\alpha) Z \left(\frac{K}{L}\right)^{\alpha}. \tag{7}$$

Because the technology exhibits constant returns to scale, the firm earns zero profits in equilibrium, and factor prices depend only on the capital–labor ratio K/L. Aggregate effective labor is supplied by employed households,

$$L = \int \mathbb{1}\{e = E\}\, y(s)\, d\Phi(a, s, \beta). \tag{8}$$

Since labor supply is exogenous, aggregate effective labor is determined by the stationary distribution of employment and productivity. Given this labor supply, the equilibrium interest rate adjusts to equate the firm's capital demand with the aggregate capital supplied by households, accounting for estate losses.

4.3 Government

The government runs a balanced-budget unemployment-insurance scheme, there is no government debt, and the budget clears period by period. Benefits are a fraction λ of mean employed labor income, $b_u = \lambda\, w \mathbb{E}[y \mid \text{employed}]$, financed by the proportional labor tax τ_l:

$$\tau_l\, w\, \bar{y}\, (1 - u) = b_u\, u \qquad \Longrightarrow \qquad \tau_l = \frac{b_u\, u}{w\, \bar{y}\, (1 - u)}, \tag{9}$$

where u is the steady-state unemployment rate and $\bar{y} = \mathbb{E}[y \mid \text{employed}]$ is mean employed efficiency units.

4.4 Stationary Competitive Equilibrium

A stationary competitive equilibrium consists of a value function $V(a, s; \beta)$, policy functions $c(a, s; \beta)$ and $a'(a, s; \beta)$, a stationary distribution $\Phi(a, s, \beta)$, aggregate capital K, factor prices (R, w), and a tax rate τ_l such that:

1. given (R, w, τ_l), the policy functions solve the household problem (2);
2. factor prices satisfy the firm's optimality conditions (7);
3. the tax rate τ_l satisfies the government budget (9);
4. factor markets clear. A fraction p_d of households die each period and a share τ_b of their terminal assets does not reach their heirs,

$$K = \left(1 - p_d\,\tau_b\right) \int a'\, d\Phi, \qquad L = \int \mathbf{1}\{e = E\}\, y(s)\, d\Phi;$$

5. the goods market clears,

$$Y = C + \delta_K K + \Delta, \qquad \Delta = p_d\,\tau_b \int a'\, d\Phi,$$

   where Δ is the value of estates absorbed outside the dynasty;
6. Φ is invariant under the policy rules and the transition laws for labor income, mortality, inheritance, and discount-factor transmission.

4.5 Functional Forms and Stochastic Processes

Preferences. Period utility is CRRA, $u(c) = \dfrac{c^{1-\gamma}}{1-\gamma}$.

Labor income. Log labor productivity is the sum of a permanent inherited component and a transitory component,

$$\ln y_{it} = \xi_i + z_{it}. \tag{10}$$

The permanent component ξ_i is fixed over the individual's life and is inherited across generations. At birth it is drawn from a three-state Markov chain conditional on the parent's ξ, with transition matrix Q_ξ (Appendix A.3 gives Q_ξ explicitly), intergenerational persistence ρ_ξ, and unconditional standard deviation σ_ξ.

The transitory component follows an AR(1),

$$z_{it} = \rho_z \, z_{i,t-1} + \varepsilon_{it}, \qquad \varepsilon_{it} \sim N(0, \sigma_z^2), \tag{11}$$

where ρ_z governs persistence and σ_z is the innovation standard deviation. The transitory productivity is not inherited. Newborn households draw z from its stationary distribution. For surviving households, productivity evolves in every period, including during unemployment spells, so a household that returns to employment earns at its current productivity level.

Employment. Employment status follows a two-state Markov chain

$$P_e = \begin{bmatrix} p_{uu} & p_{ue} \\ p_{eu} & p_{ee} \end{bmatrix}, \qquad p_{eu} = 1 - p_{ee}, \quad p_{ue} = 1 - p_{uu}, \tag{12}$$

independently of productivity. Employed households earn labor income $wy(s)$, while unemployed households receive the benefit b_u.

Discount-factor transmission. At birth, descendants draw their discount-factor type according to

$$P_\beta = \begin{bmatrix} p_{\beta\beta} & 1 - p_{\beta\beta} \\ 1 - p_{\beta\beta} & p_{\beta\beta} \end{bmatrix}, \tag{13}$$

where $p_{\beta\beta}$ governs the intergenerational persistence of patience. The symmetric transition matrix implies equal stationary shares of the two discount-factor types.

HtM status in the model. Consistent with the empirical definition, a household is classified as HtM when its beginning-of-period assets are below a fraction of its income:

$$\mathrm{HtM}_t = \mathbf{1}\{a_t < \kappa \, n(s_t)\}, \qquad \kappa = \tfrac{1}{26}. \tag{14}$$

5 Calibration and Estimation

I divide the parameters into three groups: parameters calibrated externally, parameters estimated from the PSID in a first stage, and four structural parameters $(\tau_b, p_{\beta\beta}, \sigma_\beta, \bar{\beta})$ estimated within the model.

5.1 External Calibration

Table 2 reports externally calibrated parameters. I use standard values for production and preferences, impose no unsecured borrowing, calibrate mortality to the PSID, and set the UI replacement rate to 0.30 following Kaplan and Violante (2022).

Table 2. Externally Calibrated Parameters

Parameter	Description	Value	Source
α	Capital share	0.33	Standard macro calibration
δ_K	Depreciation rate	0.10	Standard macro calibration
γ	CRRA coefficient	2.00	Standard value in the literature
$\bar{a}$	Borrowing limit	0	No unsecured borrowing
p_d	Death probability	0.0225	Average adult lifespan in the PSID
λ	UI replacement rate	0.30	Kaplan and Violante (2022)

Notes: These parameters are fixed prior to the structural estimation.

5.2 First-Stage Estimation

I estimate the parameters governing the exogenous labor-income and employment processes using the PSID. These parameters are held fixed in the structural estimation of Section 5.3.

Parameters and moments. To estimate the earnings process specified in Section 4.5, I model observed residual log household labor income as the sum of permanent earnings capacity, a persistent earnings component, and classical measurement error:

$$r_{it} = \xi_i + z_{it} + m_{it}, \qquad z_{it} = \rho_z z_{i,t-1} + \varepsilon_{it}, \tag{15}$$

where r_{it} denotes log household labor income net of age and year effects, and m_{it} is measurement error with standard deviation σ_m.

The first-stage parameter vector contains seven parameters:

$$\theta_1 = \left(\underbrace{\rho_z, \sigma_z, \sigma_\xi, \sigma_m, \rho_\xi}_{\text{Earnings process}}, \underbrace{p_{uu}, p_{eu}}_{\text{Employment process}} \right)', \tag{16}$$

where p_{uu} is the probability of remaining unemployed and p_{eu} is the probability of transitioning from employment to unemployment. In the baseline, I maintain a fixed value of ρ_z and estimate the remaining six parameters.

I identify these parameters using ten empirical moments:

$$\hat{m}_1 = \left(\hat{\gamma}(0), \hat{\gamma}(2), \ldots, \hat{\gamma}(12), \widehat{\mathrm{Cov}}(\bar{r}_c, \bar{r}_p), \hat{f}_{uu}, \hat{f}_{eu} \right)'. \tag{17}$$

The first seven moments are within-person autocovariances of residual income, $\gamma(d) = \mathrm{Cov}(r_{it}, r_{i,t-d})$, computed from employed household heads aged 30–55 over the 1994–2023 PSID waves. The eighth is the covariance of average residual income between children and their fathers, computed from 3,665 linked pairs over the 1984–2023 waves. The final two moments are the weighted frequencies of remaining unemployed and transitioning from employment to unemployment between consecutive observations.

The theoretical counterparts are

$$\gamma(0) = \sigma_\xi^2 + \sigma_{z,\infty}^2 + \sigma_m^2, \tag{18}$$

$$\gamma(d) = \sigma_\xi^2 + \rho_z^d \sigma_{z,\infty}^2, \qquad d \in \{2, 4, \ldots, 12\}, \tag{19}$$

$$\mathrm{Cov}(\bar{r}_c, \bar{r}_p) = \rho_\xi \sigma_\xi^2, \tag{20}$$

$$f_{uu} = p_{uu}, \qquad f_{eu} = p_{eu}, \tag{21}$$

where $\sigma_{z,\infty}^2 = \sigma_z^2/(1-\rho_z^2)$. The cross-generational expression assumes that transitory shocks and measurement errors are uncorrelated across generations.

Estimator. Holding ρ_z fixed, I estimate the variances of permanent earnings capacity and persistent earnings shocks by minimum distance, matching the six positive-lag autocovariances with inverse-bootstrap-variance weights. Measurement-error variance is recovered from the contemporaneous variance, while intergenerational persistence in permanent earnings capacity is identified from the parent–child covariance of average residual earnings. Employment transition probabilities are estimated directly from weighted transition frequencies. Standard errors are based on 300 bootstrap replications and are conditional on the maintained value of ρ_z.

For the restricted and one-generation benchmarks, I estimate a single-component AR(1)

earnings process without permanent earnings capacity. Following Kaplan and Violante (2014), persistence, innovation variance, and measurement-error variance are estimated using residual earnings autocovariances at lags zero, two, and six. Table 3 reports the estimates and fitted moments for both specifications.

Persistence of the transitory component. Permanent earnings heterogeneity and highly persistent transitory shocks are difficult to distinguish over the twelve-year lag window available in the PSID. The permanent component determines the long-run level of the autocovariance profile, while ρ_z governs its rate of decay. Unrestricted minimum-distance estimation yields $\hat{\rho}_z = 0.92$, but this estimate can be sensitive to the moments used. I therefore maintain $\rho_z = 0.92$ in the baseline and examine alternative values in the robustness analysis.

Appendix Table A4 examines the sensitivity of the first-stage estimates to alternative values of ρ_z, while Section 7 shows that the main structural and welfare results remain robust to these alternatives as the welfare gain is 0.43–0.47 percent across ρ_z.

Estimates and fit. Table 3 reports the parameter estimates and compares the empirical moments with their theoretical counterparts. The permanent earnings component has a standard deviation of 0.530 and a parent–child correlation of 0.556. The persistent component has an innovation standard deviation of 0.141 and an implied shock half-life of 8.3 years. Permanent earnings capacity accounts for 68 percent of persistent earnings variance.

The employment estimates are $p_{uu} = 0.350$ and $p_{eu} = 0.044$, implying a stationary unemployment rate of 6.4 percent.

The single-component AR(1) specification implies substantially greater persistence, with $\rho_z = 0.979$ and a shock half-life of 32.2 years. Both processes imply the same persistent variance, 0.411, but the single-component process attributes it entirely to shocks and therefore implies no parent–child covariance, against 0.156 in the data.

Table 3. First-Stage Estimates and Moment Fit

Panel A. Parameter Estimates					
Parameter	Description	Baseline	S.E.	AR(1)	S.E.
ρ_z	Persistence of z	0.920	calibrated	0.979	(0.003)
σ_z	Innovation s.d.	0.141	(0.010)	0.131	(0.011)
σ_ξ	Permanent earnings s.d.	0.530	(0.020)	—	—
σ_m	Measurement-error s.d.	0.333	(0.009)	0.334	(0.008)
ρ_ξ	Parent–child correlation	0.556	(0.061)	—	—
p_{uu}	Remain unemployed	0.350	(0.004)	0.350	(0.004)
p_{eu}	Enter unemployment	0.044	(0.001)	0.044	(0.001)
Implied					
	Persistent earnings variance	0.411		0.411	
	Permanent share of persistent variance	0.684		0	
	Shock half-life (years)	8.3		32.2	
Panel B. Empirical and Fitted Moments					
Moment		Data	S.E.	Baseline	AR(1)
$\gamma(0)$		0.522	(0.009)	0.522	0.522
$\gamma(2)$		0.393	(0.010)	0.391	0.393
$\gamma(4)$		0.369	(0.012)	0.374	0.377
$\gamma(6)$		0.361	(0.012)	0.360	0.361
$\gamma(8)$		0.350	(0.014)	0.348	0.346
$\gamma(10)$		0.344	(0.015)	0.338	0.331
$\gamma(12)$		0.323	(0.015)	0.329	0.317
$\mathrm{Cov}(\bar{r}_c, \bar{r}_p)$		0.156	(0.014)	0.156	0
f_{uu}		0.350	(0.004)	0.350	0.350
f_{eu}		0.044	(0.001)	0.044	0.044

Notes: Bootstrap standard errors in parentheses. Baseline standard errors are conditional on the calibrated $\rho_z = 0.92$. The baseline fits the six positive-lag autocovariances by minimum distance and matches $\gamma(0)$, the parent–child covariance, and the employment frequencies exactly. The one-component AR(1) omits permanent earnings capacity.

5.3 Structural Estimation

The heart of the estimation focuses on four structural parameters. I jointly estimate $(\tau_b, p_{\beta\beta}, \sigma_\beta, \bar{\beta})$ by SMM using four moments: the bequest-to-income ratio, parent–child HtM correlation, wealth Gini, and mean wealth-to-income ratio. These moments primarily discipline the strength of inherited wealth, the persistence of patience, the dispersion in patience, and average patience, respectively. Appendix A.5 describes estimation and inference.

Table 4 reports the model estimates alongside a restricted economy that keeps the dy-

nastic structure but has a single discount factor, and earnings follow an AR(1) process with no inherited permanent component. This shows that an Aiyagari model with generational linkages alone is not enough. Without heterogeneity in patience and in inherited earnings capacity, the model reproduces only half the intergenerational persistence in HtM status and four-fifths of the observed wealth concentration.

Table 4. Structural Estimates and Model Fit

Panel A. Structural Parameters

Parameter	Description	Estimated	S.E.	Restricted	S.E.
τ_b	Bequest wedge	0.381	(0.080)	0.460	(0.071)
$p_{\beta\beta}$	Persistence of discount-factor type	0.724	(0.071)	—	—
σ_β	Dispersion in discount factors	0.060	(0.011)	—	—
$\bar{\beta}$	Center of the discount-factor distribution	0.908	(0.011)	0.950	(0.003)
β_L	Impatient type ($\bar{\beta} - \sigma_\beta$)	0.848	(0.022)	—	—
β_H	Patient type ($\bar{\beta} + \sigma_\beta$)	0.967	(0.004)	—	—

Panel B. Targeted Moments: Data and Model Fit

Moment	Data	Data s.e.	Estimated	Restricted	t-stat (restr.)
Bequest-to-income ratio	2.554	0.330	2.554	2.554	+0.0
Parent–child HtM correlation	0.447	0.014	0.447	0.225	−16.3
Wealth Gini coefficient	0.728	0.013	0.728	0.621	−8.5
Mean wealth-to-income ratio	2.990	0.062	2.990	2.990	+0.0

Panel C. Untargeted Moments

Moment	Data	Data s.e.	Estimated	Restricted	Source
HtM share	0.434	(0.006)	0.179	0.064	PSID
Top 10% wealth share	0.604	(0.019)	0.539	0.443	PSID

Notes: The estimated model's four parameters are estimated jointly by SMM on the four moments of Panel B, with influence-function standard errors. The restricted model retains mortality, accidental bequests and dynastic replacement but imposes a single discount factor and a one-component income process with $\rho_z = 0.979$. The restricted model is estimated on the bequest-to-income and mean wealth-to-income moments, and the parent–child correlation and the wealth Gini are untargeted.

6 Mechanism and Counterfactuals

I use the model for three exercises. First, I decompose the contribution of each dynastic channel to intergenerational persistence in HtM status by shutting down each channel in turn. Second, I evaluate welfare under more generous unemployment insurance and compare how the stationary and fixed-population criteria respond across economies. Third, I study how dynastic heterogeneity shapes the economy's response to an aggregate productivity shock.

6.1 Channel Decomposition

I quantify the contribution of the three dynastic channels by shutting down inherited wealth, earnings capacity, and patience separately and in combination, re-solving the equilibrium in each case. In the no-bequest economy the heir's initial assets are an independent draw from the ergodic asset distribution, scaled by the same factor $1 - \tau_b$ that applies to a genuine inheritance. In the no-income economy $\rho_\xi = 0$, so the permanent component of earnings capacity is drawn independently of the parent's while its grid and unconditional variance are unchanged. In the no-preference economy $p_{\beta\beta} = 0.5$, so the discount-factor type is a coin flip. I summarize their contributions using a Shapley decomposition, which accounts for interactions among the three mechanisms. Appendix A.6 reports the full set of counterfactual economies underlying the decomposition, together with the associated Shapley contributions.

Figure 4. Shapley decomposition: correlation between parent and child HtM rates



Figure 4 shows the decomposition. Inherited wealth accounts for 61 percent of the parent–child HtM correlation, inherited patience for 21 percent, and inherited earnings capacity for 18 percent. When all three channels are removed, the correlation is essentially zero, implying that all parent–child HtM persistence in the model operates through these channels.

6.2 Unemployment Insurance and Welfare

I next raise the UI replacement rate from 0.30 to 0.35 and compare the estimated model with a one-generation benchmark. The benchmark model is a one-generation incomplete-markets economy with the same unemployment-insurance system (A). A has no mortality or bequests, a single discount factor, and a one-component income process estimated from the same residuals as the baseline process[3]. Model A is calibrated to have nearly identical equilibrium interest rates and capital–labor ratios under the baseline policy.

I report welfare as consumption-equivalent variation under two evaluations. The stationary comparison evaluates each policy at its own stationary distribution, so it incorporates both the change in the value of a given household state and the change in the long-run distribution of states. The fixed-distribution comparison instead evaluates both policy regimes using the same state weights, given by each economy's stationary distribution under the initial replacement rate, and therefore measures how the baseline distribution values the alternative permanent policy regime. Appendix A.7 gives formal definitions and additional welfare results.

Table 5. Welfare Gains from Higher Unemployment Insurance

Model	Stationary	Fixed distribution
B: Heterogeneous β, dynastic	+0.450	+0.505
A: Homogeneous β, one-generation	+0.041	+0.377

Consumption-equivalent welfare gains, in percent, from raising the unemployment-insurance replacement rate from 0.30 to 0.35.

Both economies value the reform positively, but the stationary-equilibrium gains differ significantly. Welfare rises by 0.450 percent of consumption in the estimated model and only 0.041 percent in the one-generation benchmark. The small gain in model A is consistent with quantitative general-equilibrium studies in which precautionary saving and equilibrium

[3]Model A differs from the restricted model in Section 5.3. The restricted model retains mortality, bequests, and dynastic replacement but imposes a single discount factor and no inherited earnings capacity. Model A instead eliminates intergenerational linkages entirely with no mortality or bequests. Both use the single-component AR(1) earnings process.

responses limit the welfare gains from additional unemployment insurance (Young, 2004; Mukoyama, 2013).

The fixed-distribution comparison substantially narrows the gap between the estimated and one-generation economies. Holding the initial distribution of household states fixed, the welfare gains are 0.505 percent in the estimated model and 0.377 percent in model A. Thus, the difference in stationary-equilibrium welfare is not primarily due to a large difference in how the baseline distributions value the higher-insurance regime. Instead, allowing the stationary distribution to adjust reduces the welfare gain much more strongly in the one-generation economy than in the estimated model.

6.3 Short Run Path of the Economy

Finally, I study a one-time, unexpected one percent increase in aggregate TFP, decaying at rate 0.9. I compare the same two economies as in the UI exercise.

Figure 5. Impulse responses to a one-percent TFP shock across models

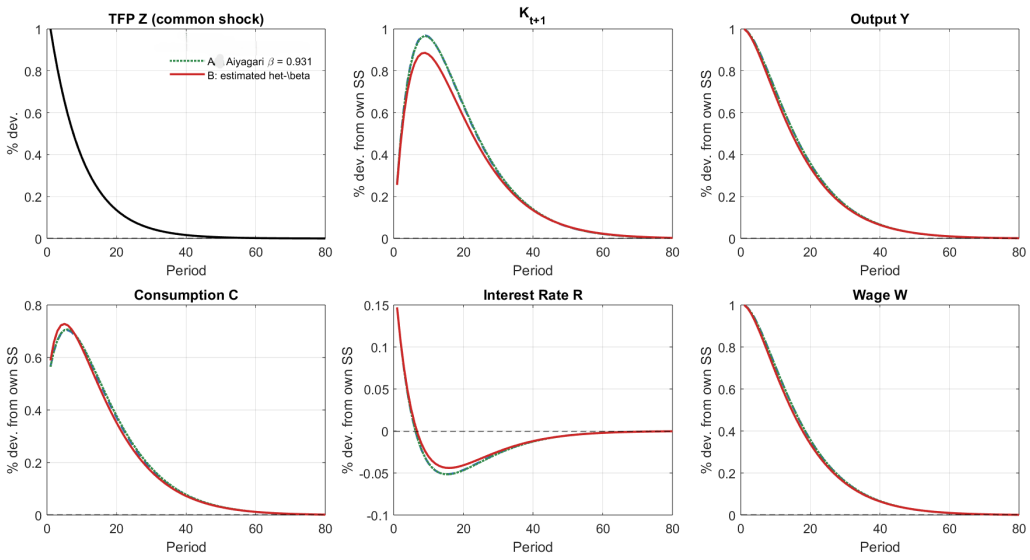
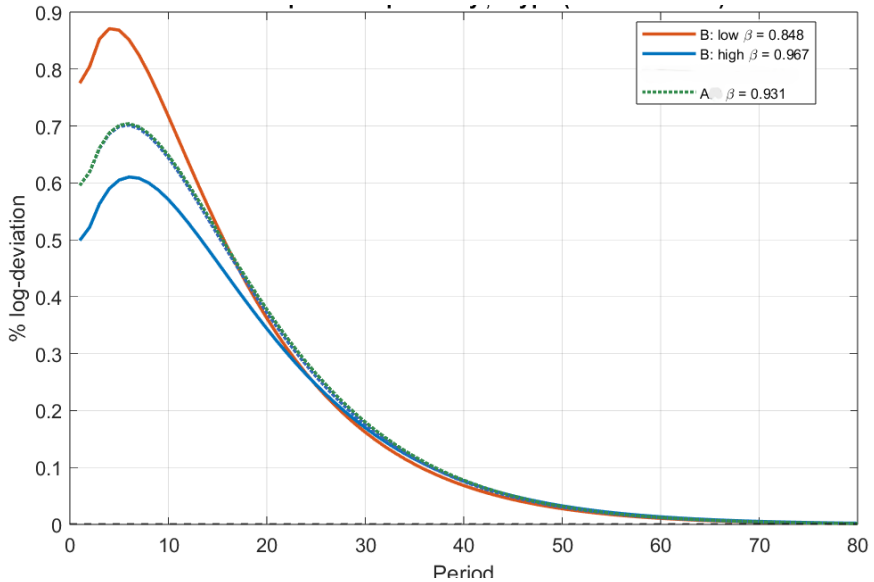


Figure 5 compares the aggregate responses across the two economies. Because capital is predetermined and labor supply is identical, output, wages, and returns move similarly on impact. The main difference is how households divide the income gain between consumption and saving. Aggregate consumption rises by 0.589 percent on impact in the estimated model, compared with 0.568 percent in the one-generation benchmark. Correspondingly, next-period capital rises slightly less in the estimated model, by 0.256 percent against 0.261 percent. These differences are quantitatively modest as the largest difference in the impact consumption response is only 0.023 percentage points.

The small aggregate differences, however, mask substantial heterogeneity across households. Figure 6 decomposes the consumption response in the estimated model by discount factor types. The impatient type increases consumption by 0.732 percent against 0.470 percent for the patient type. Consumption responses also differ by HtM status. HtM consumption rises by 1.164 percent on impact, nearly twice the response of non-HtM households.

Figure 6. Impulse responses to a one percent TFP shock by β types



The main implication is therefore distributional rather than aggregate. Because constrained households hold little wealth and account for a small share of aggregate resources, their much larger individual responses translate into only modest differences in aggregate consumption. Similar aggregate dynamics can conceal substantially different exposure to aggregate shocks across households.

7 Robustness

The unemployment-insurance result is the main policy implication of the model. Raising the unemployment-insurance replacement rate from 0.30 to 0.35 increases long-run welfare by 0.45 percent of consumption in the estimated model. This section asks whether this result depends on the estimated parameters, on the model's understatement of the HtM share, on the maintained persistence of the transitory income component, on the absence of a bequest motive, or on the size of the policy change. Every exercise re-solves general equilibrium.

7.1 Estimated parameters

I begin with eight local perturbations of the four parameters estimated by SMM. Table 6 shows the UI exercise results of perturbing each structural parameter by one standard error, holding the other three at their estimates.

Table 6. Welfare Gain Under Perturbed Structural Parameters

Scenario	Value	Long-run CEV (%)	HtM share	R^*
Baseline	—	+0.450	0.179	1.0546
σ_β high	0.071	+0.454	0.192	1.0430
σ_β low	0.048	+0.427	0.163	1.0663
$\bar{\beta}$ high	0.919	+0.443	0.170	1.0433
$\bar{\beta}$ low	0.897	+0.459	0.189	1.0661
τ_b high	0.461	+0.456	0.180	1.0555
τ_b low	0.301	+0.443	0.179	1.0536
$p_{\beta\beta}$ high	0.794	+0.460	0.181	1.0541
$p_{\beta\beta}$ low	0.653	+0.439	0.177	1.0550

The welfare gain lies between 0.43 and 0.46 percent across all eight perturbations, a range of only 0.03 percentage points around the baseline estimate of 0.45 percent. I therefore conclude that the positive welfare effect of the UI reform is not driven by any particular structural parameter estimate and is locally robust to the estimated uncertainty in all four parameters.

7.2 Matching the empirical HtM share

The baseline model generates an HtM share of 0.179, compared with 0.434 in the data. To assess whether this discrepancy matters for the results, I lower the impatient discount factor, holding the patient type at its estimate, until the model matches the empirical HtM share. Reducing β_L from 0.848 to 0.515 raises the HtM share to 0.434. Because the patient type holds nearly all of the capital, aggregate outcomes are little changed. The equilibrium interest rate rises from 1.0546 to 1.0560 and the capital–output ratio falls from 2.135 to 2.115. Wealth concentration rises, with the Gini increasing from 0.728 to 0.772.

Holding the remaining parameters fixed, the lower β_L also raises the parent–child HtM correlation, from 0.447 to 0.537. With a wider gap between the two types, HtM status becomes closely tied to discount-factor type, and types are persistent across generations. I therefore re-fit the persistence of types so that the economy matches both the HtM share and the parent–child correlation. This requires $p_{\beta\beta} = 0.653$, exactly one standard error below its estimate of 0.724. The bequest-to-income ratio (2.541) and the wealth-to-income

ratio (2.962) remain within half a standard error of their targets, while the Gini rises to 0.777.

The welfare result is robust to matching the HtM share. In the re-fitted economy, raising the UI replacement rate from 0.30 to 0.35 increases long-run welfare by 0.439 percent of consumption, compared with 0.450 percent in the baseline. Holding $p_{\beta\beta}$ at its baseline value instead gives a gain of 0.443 percent.

Matching the HtM share does, however, change how intergenerational persistence is divided across channels. In the re-fitted economy, a Shapley decomposition attributes 53 percent of the parent–child correlation to inherited wealth, 43 percent to inherited patience, and 4 percent to inherited earnings capacity (Table 7). The contribution of inherited wealth is nearly unchanged in levels, at 0.237 against 0.274 in the baseline, and it remains the largest single channel. What changes is the division of the remaining persistence. When many households are impatient enough to be constrained regardless of their earnings, inherited patience rather than inherited earnings capacity accounts for it.

Table 7. Shapley Decomposition When the Model Matches the HtM Share

	Baseline	β_L lowered, $p_{\beta\beta}$ fixed	β_L lowered, $p_{\beta\beta}$ re-fit
HtM share	0.179	0.435	0.434
$p_{\beta\beta}$	0.724	0.724	0.653
Parent–child HtM correlation	0.447	0.537	0.447
Shapley contributions (share of correlation)			
Inherited wealth	0.274 (61%)	0.239 (44%)	0.237 (53%)
Inherited earnings capacity	0.081 (18%)	0.019 (3%)	0.018 (4%)
Inherited patience	0.094 (21%)	0.281 (52%)	0.193 (43%)
Wealth Gini	0.728	0.772	0.777

Notes: The second and third columns lower β_L to 0.515, holding β_H at its estimate, so that the stationary HtM share matches the data. The third column also re-fits $p_{\beta\beta}$ to match the parent–child HtM correlation. Every scenario is re-solved in general equilibrium.

7.3 Persistence of the transitory income component

The baseline specification fixes the persistence of the transitory earnings component at $\rho_z = 0.92$. I assess sensitivity to this maintained assumption using $\rho_z \in \{0.86, 0.88, 0.90, 0.94, 0.96\}$ with the corresponding (σ_ξ, ρ_ξ) from Table A4. For each value, I re-estimate the first-stage income-process parameters affected by ρ_z, re-estimate the structural parameters by SMM, resolve the general equilibrium, and recompute the UI counterfactual. The comparison

therefore asks whether the results depend on which income decompositions the first stage selects.

Table 8. Estimates and Welfare Gain Across Maintained Values of ρ_z

	First stage		*Structural estimates*				*Model*	
ρ_z	σ_ξ	ρ_ξ	τ_b	$p_{\beta\beta}$	σ_β	$\bar{\beta}$	HtM share	CEV (%)
0.86	0.561	0.496	0.389	0.705	0.076	0.894	0.198	+0.470
0.88	0.555	0.509	0.387	0.709	0.071	0.898	0.189	+0.466
0.90	0.545	0.527	0.384	0.713	0.065	0.903	0.190	+0.457
0.92	0.530	0.556	0.381	0.724	0.060	0.908	0.179	+0.450
0.94	0.505	0.614	0.377	0.732	0.053	0.913	0.165	+0.436
0.96	0.449	0.776	0.367	0.777	0.046	0.919	0.146	+0.426

Notes: Each row re-estimates all four structural parameters at the stated maintained ρ_z, using the first-stage (σ_ξ, ρ_ξ) that the same lag window delivers under that value, and recomputes the long-run consumption-equivalent gain from raising the replacement rate from 0.30 to 0.35. The row $\rho_z = 0.92$ is the baseline.

As table 8 shows, the welfare gain lies between 0.43 and 0.47 percent, a spread of 0.04 percentage points around a baseline of 0.45. The structural estimates vary monotonically, though the changes are quantitatively small. On one hand, the bequest wedge falls by six percent across the range and the center of the discount-factor distribution rises by three.

On the other hand, the dispersion in discount factors falls from 0.076 to 0.046. As ρ_z rises, the more persistent transitory component accounts for a larger share of long-run earnings variation, reducing the variance attributed to the permanent component. Given the observed parent–child covariance in earnings, the smaller permanent variance implies a higher estimated intergenerational correlation ρ_ξ, which rises from 0.50 to 0.78. The greater persistence of inherited earnings capacity then accounts for more of the targeted parent–child HtM correlation, reducing the amount of discount-factor dispersion required to match it. The resulting decline in σ_β leaves fewer households near the liquidity constraint, with the model's HtM share falling from 0.198 to 0.146, accompanied by a slightly smaller welfare gain.

7.4 A bequest motive

The baseline treats bequests as accidental. Households do not value what they leave, and the transfer arises only because some die holding assets. De Nardi (2004) shows that voluntary transfers concentrate wealth differently from accidental ones. I therefore add a warm-glow

term over the amount the heir receives,

$$\phi(b) = \phi_1 \frac{(1 + b/\phi_2)^{1-\gamma}}{1-\gamma}, \qquad b = (1-\tau_b)a',$$

so the household problem gains the term $\beta p_d \, \phi\big((1-\tau_b)a'\big)$ inside the maximization. The warm glow enters the value function as well as the Euler. I hold ϕ_2 at 11.6 and set $\phi_1 = 9.5$, which corresponds to De Nardi's estimate under this parametrization, and re-estimate all four structural parameters.

Table 9. Estimates and Welfare Gain With a Bequest Motive

	Accidental bequests	Warm glow, $\phi_1 = 9.5$
Bequest wedge τ_b	0.381 (0.080)	0.327 (0.087)
Type persistence $p_{\beta\beta}$	0.724 (0.071)	0.949 (0.102)
Dispersion σ_β	0.060 (0.011)	0.028 (0.007)
Center $\bar{\beta}$	0.908 (0.011)	0.922 (0.006)
Impatient type β_L	0.848	0.894
Patient type β_H	0.967	0.951
HtM share	0.179	0.126
Top 10 percent wealth share	0.539	0.563
Long-run CEV (percent)	+0.450	+0.237
value effect	+0.079	+0.071
composition effect	−0.009	−0.034

Notes: Influence-function standard errors in parentheses. Both columns match the four targeted moments exactly. The value and composition effects are in welfare levels and decompose the stationary comparison as in Appendix A.7.

The motive substitutes for discount-factor heterogeneity rather than adding to it. Dispersion in patience halves, from 0.060 to 0.028, because the bequest motive generates wealth concentration on its own and less dispersion is needed to match the wealth Gini. The parent–child correlation must still be matched, however, and with half the dispersion available the persistence of discount-factor type rises almost to one, from 0.72 to 0.95. Wealth concentration improves slightly, the top-decile share rising from 0.539 to 0.563 against 0.605 in the data, while the HtM share falls from 0.179 to 0.126, moving the model further from its empirical counterpart.

Overall, the welfare gain falls from 0.450 to 0.237 percent of consumption. The reform's direct value is little changed, holding the pre-reform distribution fixed, the welfare gain is +0.071, compared with +0.079 in the baseline. Instead, the difference comes from the long-run distributional response, whose welfare effect becomes substantially more negative, from −0.009 to −0.034. More generous unemployment insurance reduces precautionary

saving and therefore the assets households carry into death. Under accidental bequests, this matters primarily because heirs begin with less wealth. With a warm-glow bequest motive, lower asset holdings are also costly because households value the estates they leave. The reform-induced shift toward lower wealth therefore carries a larger welfare cost, although the decomposition reflects both changes in the stationary distribution and changes in the value attached to those states. Scaling the bequest term together with consumption when computing the compensating variation yields a welfare gain of 0.234 percent, confirming that the result is not driven by the treatment of bequest utility in the CEV calculation.

7.5 Size of the policy change

Finally, I vary the replacement rate over a substantially wider range rather than considering only the local increase from 0.30 to 0.35. Results show that welfare is lower when replacement is reduced below the baseline and rises monotonically over the range above it. For example, the consumption-equivalent gain is 0.450 percent at a replacement rate of 0.35, 0.877 percent at 0.40, 1.682 percent at 0.50, and 3.135 percent at 0.70. Conversely, reducing the replacement rate to 0.25 lowers welfare by 0.456 percent and reducing it to 0.20 lowers welfare by 0.919 percent. The baseline reform is therefore not a knife-edge comparison around 0.30 as the estimated economy consistently values additional unemployment insurance over the range considered.

8 Conclusion

This paper shows that HtM status is persistent across generations. Children raised in households that spend more time HtM are substantially more likely to remain HtM themselves as adults, and this relationship persists after accounting for observable parental characteristics. The quantitative model identifies inherited wealth as the largest source of this persistence, accounting for 61 percent of the intergenerational correlation in HtM status, and it remains the largest channel when the model matches the empirical HtM share. Wealth alone, however, is not enough. Inherited earnings capacity and persistent differences in saving behavior together account for the remaining two-fifths of persistence. Financial vulnerability is therefore not simply the consequence of low current resources or temporary shocks. It partly reflects economic circumstances and behavior that persist across generations.

This dynastic persistence also matters for the evaluation of macroeconomic policy. Raising unemployment-insurance generosity produces a substantially larger stationary-equilibrium welfare gain in the estimated dynastic economy than in a one-generation incomplete-markets benchmark, despite sharing the same equilibrium interest rate and capital–labor ratio. This

difference narrows considerably when both policy regimes are evaluated using the initial distribution of household states, indicating that the endogenous response of the long-run distribution is an important part of the stationary-equilibrium comparison. The same heterogeneity shapes responses to aggregate shocks in ways that aggregate consumption dynamics do not reveal. HtM households adjust their consumption nearly twice as much as households with greater liquidity.

More broadly, the results suggest that financial vulnerability is partly a dynastic outcome rather than simply a reflection of a household's current balance sheet. Wealth, earnings capacity, and saving behavior transmitted across generations shape which households remain close to the liquidity constraint and therefore the distributional consequences of social insurance and aggregate shocks. Accounting for these intergenerational forces can consequently matter for both the distributional and welfare implications of macroeconomic policy. Future work could extend the framework to distinguish liquid from illiquid assets, allow earnings risk as well as earnings capacity to be inherited (Halvorsen et al., 2022), and incorporate richer life-cycle dynamics, allowing the persistence of financial vulnerability to vary over the life cycle and across different forms of wealth.

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A Appendix

A.1 Data, Sample Construction, and Measurement

PSID sample and genealogical links The baseline lifetime sample restricts both generations to ages 30–65, which reduces contamination from schooling and early household formation while retaining the economically relevant portion of working life. However, the income-process estimation uses a narrower age range, 30–55, because earnings dynamics become increasingly affected by retirement and endogenous labor-force withdrawal after the mid-fifties. The within-person covariance block uses the biennial 1994–2023 waves; the parent–child covariance block uses the full 1984–2023 window to preserve cohort coverage and the number of linked pairs.

Income, wealth, and balance-sheet variables Household income is annual labor income of the household head, expressed in 2009 dollars. Liquid wealth is the sum of checking and savings account balances and certificates of deposit. Total net wealth includes liquid financial assets, other financial assets, business wealth, housing equity, vehicles, and other real assets net of debts available in the relevant PSID wealth module. Illiquid wealth is defined residually as total net wealth less liquid wealth. The construction therefore separates the resource that can be used immediately for consumption smoothing from wealth that is costly or slow to access.

For regressions that include wealth I use the inverse hyperbolic sine transformation,

$$\mathrm{IHS}(x) = \log\Big(x + \sqrt{x^2 + 1}\Big),$$

which is defined at zero and for negative values.

Intergenerational transfers and the bequest moment The PSID records inheritances and inter vivos transfers received at different ages. The model instead delivers a single transfer when a descendant is born. I map observed transfers into a common present-value object before comparing them with the model. Let T_{ij} denote transfer j received by child i at age a_{ij}, let a_0 denote the model entry age, and let r be the annual real discount rate used for this accounting exercise. The entry-age value is

$$B_i = \sum_j \frac{T_{ij}}{(1+r)^{a_{ij}-a_0}}.$$

A.2 HtM Definitions and Descriptive Evidence

Baseline definition Let L_{it} be liquid wealth, W_{it} total net wealth, I_{it} annual income, and $Q_{it} = W_{it} - L_{it}$ illiquid wealth. A household is HtM if

$$H_{it} = \mathbf{1}\{L_{it} < -0.165\, I_{it}\} + \mathbf{1}\left\{0 \leq L_{it} \leq \tfrac{I_{it}}{26}\right\}, \tag{A.1}$$

where the two events are mutually exclusive. Among HtM households,

$$H^P_{it} = H_{it}\, \mathbf{1}\left\{Q_{it} \leq \tfrac{I_{it}}{6}\right\}, \tag{A.2}$$

$$H^W_{it} = H_{it}\, \mathbf{1}\left\{Q_{it} > \tfrac{I_{it}}{6}\right\}. \tag{A.3}$$

The negative-liquid-wealth threshold follows the Kaplan–Violante–Weidner convention, $I/26$ is two weeks of annual income and $I/6$ two months.

Starting from liquidity matters here. A classification that first labels households using total net worth and only then identifies a liquid subgroup mixes two distinct concepts. A household can have low net worth without being close to the liquid-resource margin relevant for short-run consumption smoothing. The baseline definition applies the same liquidity screen to both poor and wealthy HtM households and uses illiquid wealth only to partition the constrained group.

Separately, the model imposes $\bar{a} = 0$ and so has no households with negative assets. Observations with negative net worth are therefore excluded when constructing the moments the model targets. That restriction removes 11.6 percent of working-age household-years and lowers the HtM share from the 46.9 percent reported in Section 3 to 43.4.

Alternative definitions I compare the baseline classification with two alternatives. The net-worth criterion used by Aguiar et al. (2025) classifies a household as HtM when total net worth is less than two months of earnings. The Kaplan et al. (2014) classification uses liquid resources to identify poor and wealthy HtM households but classifies households as wealthy HtM if they have any illiquid wealth. Table A1 shows the correlations between definitions.

Summary statistics by HtM status Table A2 reports characteristics by group. Poor HtM households are younger, have much lower income, and are more likely to carry high liquid debt. Wealthy HtM households also hold little liquid wealth but have substantially higher net worth, consistent with illiquid assets that do not provide immediate self-insurance.

Table A1. Correlation across alternative HtM definitions

	Baseline	KVW	Aguiar/net worth
Panel A. Poor HtM / net-worth HtM			
Baseline poor HtM	1.0000		
KVW poor HtM	0.7975	1.0000	
Aguiar/net-worth HtM	0.8526	0.6932	1.0000
Panel B. Wealthy/liquid HtM			
Baseline wealthy HtM	1.0000		
KVW wealthy HtM	0.8338	1.0000	
KVW liquid HtM	0.9916	0.8409	1.0000
Panel C. Overall HtM			
Baseline HtM	1.0000		
KVW liquid HtM	1.0000	1.0000	
Aguiar broad HtM	0.8808	0.8808	1.0000

Within-generation persistence Following Aguiar et al. (2025), table A3 shows that HtM status does not consist primarily of short-lived liquidity episodes. Both poor and wealthy HtM households are far more likely to remain HtM than households that begin outside the HtM state.

A.3 First-Stage Earnings and Employment Processes

Sample The earnings sample consists of household heads aged 30–55 with real annual labor income above $2,000. Log earnings are winsorized at the first and ninety-ninth percentiles. I regress log labor income on age and year fixed effects using PSID family weights and cluster by individual. Earnings-process moments use employed observations because unemployed households receive a fixed replacement benefit in the model.

The within-person autocovariances use the 1994–2023 waves. The sample begins in 1994 because the employment variables used to restrict the sample to employed heads are not available in earlier waves. Because the PSID moved from annual to biennial interviews after 1997, I use only even lags, $d \in \{2, 4, \ldots, 12\}$, which are observed throughout the sample period. The cross-generation covariance uses the full 1984–2023 window to preserve cohort coverage and the number of linked parent–child pairs.

Process and identifying moments Residual earnings follow

$$r_{it} = \xi_i + z_{it} + m_{it}, \qquad z_{it} = \rho_z z_{i,t-1} + \epsilon_{it}, \qquad \epsilon_{it} \sim N(0, \sigma_z^2), \tag{A.4}$$

Table A2. Summary characteristics by HtM status

Statistic	Not HtM	Poor HtM	Wealthy HtM
Age	47.1	43.5	46.4
Income	104,054.9	45,103.8	78,172.7
Earnings	63,609.7	27,589.3	45,842.2
High liquid debt (percent)	13.5	21.9	12.4
Sample share (percent)	53.1	19.8	27.1
Liquid wealth, median	14,197.7	2.6	362.8
Net worth, median	166,067.0	0.0	66,831.1

Notes: Monetary values are in 2009 dollars. High liquid debt equals one when credit-card debt, store credit, student loans, medical or legal bills, and loans from family sum to at least one month of income. Means are weighted by PSID family weights. Medians are weighted percentiles.

Table A3. Transition probabilities of HtM status in the PSID

Initial status	Poor HtM	Wealthy HtM	Not HtM
Panel A. Two-year horizon			
Poor HtM	60.63	21.60	17.77
Wealthy HtM	18.00	51.66	30.34
Not HtM	8.89	18.33	72.79
Panel B. Four-year horizon			
Poor HtM	55.40	24.29	20.32
Wealthy HtM	18.71	48.03	33.25
Not HtM	8.71	19.18	72.11

Notes: Row percentages from weighted transition matrices.

where ξ_i is fixed within life and inherited across generations, z_{it} is a transitory AR(1), and m_{it} is classical measurement error. Let $\sigma^2_{z,\infty} = \sigma^2_z/(1-\rho^2_z)$ and $\sigma^2_\xi = \mathrm{Var}(\xi_i)$. Then

$$\gamma(0) = \sigma^2_\xi + \sigma^2_{z,\infty} + \sigma^2_m, \tag{A.5}$$

$$\gamma(d) = \sigma^2_\xi + \rho^d_z \sigma^2_{z,\infty}, \qquad d \geq 1. \tag{A.6}$$

Conditional on ρ_z, the long-run level of the autocovariance profile identifies σ^2_ξ, its decay identifies $\sigma^2_{z,\infty}$, and σ^2_m follows from $\gamma(0)$.

For individual i, let $\bar{r}_i$ denote average residual earnings over observed working-age years. Independence of transitory shocks and measurement error across generations implies

$$\mathrm{Cov}(\bar{r}_c, \bar{r}_p) = \rho_\xi \sigma^2_\xi, \tag{A.7}$$

so the parent–child covariance identifies ρ_ξ conditional on the permanent variance estimated from the within-person moments.

Estimator. Given ρ_z, I estimate $(\sigma^2_\xi, \sigma^2_{z,\infty})$ by minimum distance using the six positive-lag autocovariances, $d \in \{2, 4, \ldots, 12\}$:

$$(\hat{\sigma}^2_\xi, \hat{\sigma}^2_{z,\infty}) = \arg\min_{\sigma^2_\xi, \sigma^2_{z,\infty}} \sum_{d \in \{2,4,\ldots,12\}} \omega_d \left[\hat{\gamma}(d) - \sigma^2_\xi - \rho^d_z \sigma^2_{z,\infty}\right]^2, \tag{A.8}$$

where ω_d is the inverse bootstrap variance of $\hat{\gamma}(d)$. With ρ_z fixed, the problem is linear in the two variance parameters, and the six moments provide four overidentifying restrictions. Given these estimates, measurement-error variance and intergenerational persistence are recovered from

$$\hat{\sigma}^2_m = \hat{\gamma}(0) - \hat{\sigma}^2_\xi - \hat{\sigma}^2_{z,\infty}, \qquad \hat{\rho}_\xi = \frac{\widehat{\mathrm{Cov}}(\bar{r}_c, \bar{r}_p)}{\hat{\sigma}^2_\xi}. \tag{A.9}$$

The employment transition probabilities are estimated directly from their corresponding weighted frequencies. Standard errors are computed using 300 bootstrap replications, resampling individuals for the within-person moments and dynasties for the intergenerational covariance, and are conditional on ρ_z.

For the restricted and one-generation benchmarks, I also estimate a one-component AR(1) earnings process that excludes permanent earnings capacity. Following Kaplan and Violante (2014), I use the residual earnings autocovariances at lags zero, two, and six to estimate persistence, innovation variance, and measurement-error variance. This specification attributes persistent earnings variation to a single autoregressive component rather

than distinguishing permanent earnings capacity from transitory shocks. Its estimates and fitted moments are reported alongside the baseline in Table 3.

Sensitivity to maintained ρ_z Because the permanent and highly persistent transitory components are only weakly separated over the available lag window, I assess the sensitivity of the first-stage estimates to alternative maintained values of ρ_z. Table A4 reports the resulting estimates.

Table A4. First-Stage Estimates Across Maintained Values of ρ_z

ρ_z	σ_ξ	S.E.	ρ_ξ	S.E.	Permanent share
0.86	0.561	(0.015)	0.496	(0.049)	0.750
0.88	0.555	(0.016)	0.509	(0.051)	0.737
0.90	0.545	(0.017)	0.527	(0.054)	0.717
0.92	**0.530**	**(0.020)**	**0.556**	**(0.061)**	**0.684**
0.94	0.505	(0.024)	0.614	(0.077)	0.623
0.96	0.449	(0.035)	0.776	(0.142)	0.496

Notes: Each row re-estimates the first-stage variance parameters at the stated value of ρ_z. Standard errors are based on 300 bootstrap replications and are conditional on the maintained persistence parameter. The bold row is the baseline. Section 7 re-estimates the structural model for $\rho_z \in [0.86, 0.96]$.

The one-component specification The restricted model of Table 4 and the one-generation benchmark of Section 6.2 impose a single persistent component, so the income process must be re-estimated without a fixed effect. Setting $\sigma_\xi = 0$ in (A.6) leaves $\gamma(d) = \rho_z^d \sigma_{z,\infty}^2$, which decays to zero.

With three parameters $(\rho_z, \sigma_{z,\infty}^2, \sigma_m^2)$ the system is exactly identified by three moments. I use $\gamma(0)$, $\gamma(2)$ and $\gamma(6)$, the moment set of Kaplan and Violante (2014), which in differences is $\mathbb{E}[(r_t - r_{t-d})^2] = 2\gamma(0) - 2\gamma(d)$ and is therefore a linear map of the autocovariances used above. The estimator has a closed form:

$$\hat{\rho}_z = \left[\frac{\gamma(6)}{\gamma(2)}\right]^{1/4}, \qquad \hat{\sigma}_{z,\infty}^2 = \frac{\gamma(2)}{\hat{\rho}_z^2}, \qquad \hat{\sigma}_m^2 = \gamma(0) - \hat{\sigma}_{z,\infty}^2. \tag{A.10}$$

Mapping the income process into the model The permanent component is approximated by a three-state Rouwenhorst chain. Let $p_\xi = (1+\rho_\xi)/2$. Then

$$Q_\xi = \begin{bmatrix} p_\xi^2 & 2p_\xi(1-p_\xi) & (1-p_\xi)^2 \\ p_\xi(1-p_\xi) & p_\xi^2+(1-p_\xi)^2 & p_\xi(1-p_\xi) \\ (1-p_\xi)^2 & 2p_\xi(1-p_\xi) & p_\xi^2 \end{bmatrix},$$

with stationary probabilities $(1/4, 1/2, 1/4)$ and $\mathrm{Corr}(\xi', \xi) = 2p_\xi - 1 = \rho_\xi$ exactly. With nodes $\{-\psi, 0, \psi\}$, matching $\mathrm{Var}(\xi) = \sigma_\xi^2$ requires $\psi = \sqrt{2}\,\sigma_\xi$. More generally an n-point Rouwenhorst chain requires half-width $\sqrt{n-1}\,\sigma_\xi$: the chain's stationary distribution is Binomial$(n-1, \frac{1}{2})$ with realized variance $\psi^2/(n-1)$.

The transitory AR(1) is discretized using eleven Rouwenhorst states. Newborns draw the transitory component from its stationary distribution, while the permanent component is inherited through the estimated transition matrix. During unemployment, latent productivity continues to evolve, so earnings upon re-employment reflect the current productivity state rather than a new draw.

A.4 Numerical Solution and Simulation

The household problem is solved by the endogenous-grid method on a 500-point asset grid, with the transitory earnings process discretized on 11 Rouwenhorst states. The stationary distribution is computed on a finer 1,000-point asset grid using linear lottery weights. General equilibrium is found by bisection on the gross interest rate, re-solving household policies and the invariant distribution at each candidate value. The upper bound of the search interval is the stationarity limit $1/[\beta_H(1-p_d)]$.

Stationary wealth moments are computed directly from the invariant distribution. The parent–child HtM correlation instead requires simulated genealogical links. I simulate 6,000 dynasties for 850 periods, discard the first 350, and retain the final 500. Consecutive lifetimes along a dynasty form parent–child pairs, and each lifetime is sampled every other period to mimic the biennial PSID. HtM exposure is the fraction of sampled periods for which

$$a_t < \kappa n(s_t), \qquad \kappa = \frac{1}{26}.$$

All parameter evaluations use a fixed random seed and common random numbers. Changing the simulation seed moves the parent–child HtM correlation by only 0.0018 at the estimated parameters.

A.5 Structural Estimation and Identification

Estimator Let $\theta = (\tau_b, p_{\beta\beta}, \sigma_\beta, \bar{\beta})'$ and let $m(\theta)$ contain the bequest-to-income ratio, the parent–child HtM correlation, the wealth Gini, and the mean wealth-to-income ratio. The estimator is

$$\hat{\theta} = \arg\min_{\theta \in \Theta} \left[m(\theta) - \hat{m}\right]' W \left[m(\theta) - \hat{m}\right]. \tag{A.11}$$

Because the system is exactly identified, $\hat{\theta}$ solves $m(\theta) = \hat{m}$ up to numerical tolerance. $\hat{\theta}$ is found by damped Newton on $g(\theta) = m(\theta) - \hat{m}$ rather than by minimizing (A.11).

Three of the four moments (the Gini, the mean wealth-to-income ratio and the bequest-to-income ratio) are computed from the stationary distribution. Only the parent–child correlation requires the simulated panel.

Jacobian, standard errors, and sensitivity Let $G = \partial m(\theta)/\partial \theta'$ evaluated at $\hat{\theta}$ and let Ω denote the asymptotic covariance of $\sqrt{N}(\hat{m} - m)$, so that the standard error of data moment j is $se(\hat{m}_j) = \sqrt{\Omega_{jj}/N}$. The asymptotic variance of $\hat{\theta}$ is

$$\mathrm{Var}(\hat{\theta}) = \frac{1}{N}(G'WG)^{-1}G'W\Omega WG(G'WG)^{-1},$$

which under exact identification simplifies to $N^{-1}G^{-1}\Omega(G^{-1})'$.

Due to the four moments differing in scale I also report the scaled sensitivity

$$G^s = 0.01 \cdot \mathrm{diag}(1/se(\hat{m}))\; G\; \mathrm{diag}\left(|\hat{\theta}|\right), \tag{A.12}$$

whose (j, k) entry is the change in moment j, in units of its own data standard error, produced by a one-percent change in parameter k. Diagnostics are computed on G^s rather than G, since summaries of the raw Jacobian reflect the units of the moments and parameters rather than the conditioning of the identification problem. The factor 0.01 rescales a proportional change to a one-percent change and leaves the condition number unaffected.

Panel B supports the informal assignment of moments to parameters in Section 5.3, read as a statement about which moment identifies each parameter rather than the reverse. The parent–child correlation is by far the moment most responsive to $p_{\beta\beta}$, at 0.24 standard errors per one-percent change against at most 0.02 for any other moment. The wealth Gini is the moment most responsive to σ_β relative to its own sensitivity to the other parameters. The bequest-to-income ratio is the moment most responsive to τ_b, though in absolute terms it responds far more strongly to $\bar{\beta}$; τ_b is identified because no other moment responds

Table A5. Jacobian and scaled sensitivity at $\hat{\theta}$

	τ_b	$p_{\beta\beta}$	σ_β	$\bar{\beta}$
Panel A. Raw Jacobian $G = \partial m/\partial \theta'$				
Bequest-to-income ratio	−4.392	0.182	17.748	17.339
Parent–child HtM correlation	−0.023	0.446	2.035	−0.533
Wealth Gini	0.041	−0.035	0.936	0.006
Mean wealth-to-income ratio	−0.227	0.115	19.830	19.722
Panel B. Scaled sensitivity G^s				
Bequest-to-income ratio	−0.051	0.004	0.032	0.477
Parent–child HtM correlation	−0.006	0.236	0.089	−0.353
Wealth Gini	0.012	−0.020	0.044	0.005
Mean wealth-to-income ratio	−0.014	0.013	0.189	2.865

Notes: Panel B reports the change in each moment, in units of its own data standard error, from a one-percent change in each parameter.

appreciably to it. And the mean wealth-to-income ratio is overwhelmingly the moment that identifies $\bar{\beta}$: a one-percent change in the level of patience moves it by 2.87 data standard errors, against 0.005 for the Gini, a ratio of 637.

Restricted model. The restricted model of Table 4 is exactly identified by the bequest-to-income ratio and the mean wealth-to-income ratio, and its standard errors use the same formula with the corresponding 2×2 block of Ω, which retains the covariance between the two moments. The Jacobian is computed by two-sided finite differences at the estimates, with every perturbed economy re-solved in general equilibrium. Halving the relative step from 0.02 to 0.01 changes the standard errors by less than one percent.

A.6 Channel Decomposition

Counterfactual definitions To quantify the contribution of each dynastic channel, I construct counterfactual economies that selectively shut down the intergenerational transmission of wealth, earnings capacity, and discount-factor type while preserving their unconditional distributions.

- **No wealth transmission.** The heir's initial assets are an independent draw from the ergodic asset distribution, scaled by the same factor $1 - \tau_b$ that applies to a genuine inheritance, and the assignment of transfers across heirs is randomized.

- **No earnings transmission.** $\rho_\xi = 0$, so the permanent component is drawn independently of the parent's while the ξ grid and its unconditional variance are unchanged.
- **No preference transmission.** $p_{\beta\beta} = 1/2$, so the discount-factor type is a fair coin. Both types retain equal population mass and σ_β is unchanged.

The decomposition concerns intergenerational dependence, not cross-sectional inequality so no counterfactual removes heterogeneity, only its transmission.

Shapley contributions For channel j,

$$\phi_j = \sum_{S \subseteq N \setminus \{j\}} \frac{|S|!\,(|N| - |S| - 1)!}{|N|!} \left[v(S \cup \{j\}) - v(S) \right], \tag{A.13}$$

so that $\sum_j \phi_j = v(N) - v(\varnothing)$. Applied to Table A6 this gives 0.273 for inherited wealth, 0.094 for inherited patience and 0.081 for inherited earnings capacity, summing to 0.448, or 61, 21 and 18 percent of the transmitted correlation. Severing all three leaves $v(\varnothing) = -0.001$, so the model attributes essentially all parent–child HtM persistence to the mechanisms it makes explicit rather than to residual dynastic correlation.

Table A6. Equilibria underlying the Shapley decomposition

S	Channels removed	HtM corr
N	—	0.447
$\{W, B\}$	Earnings	0.341
$\{W, Y\}$	Patience	0.338
$\{W\}$	Earnings, patience	0.253
$\{Y, B\}$	Wealth	0.155
$\{B\}$	Wealth, earnings	0.078
$\{Y\}$	Wealth, patience	0.055
$\varnothing$	All three	−0.001

Notes: $v(S)$ is the parent–child HtM correlation in the stationary equilibrium with exactly the channels in S inherited. Every economy is re-solved in general equilibrium.

Transmission versus level. Across the eight coalitions the parent–child correlation spans 0.45 while the average HtM rate spans 0.011. Dynastic transmission determines which dynasties spend much of their working lives constrained, not what fraction of a working life the average household spends constrained.

A.7 Unemployment Insurance: Welfare Definitions and Implementation

For each replacement rate r, I solve the stationary general equilibrium, including the balanced-budget labor-income tax, and obtain the associated value function $V^r(x)$ and invariant distribution $\Phi^r(x)$, where x denotes the household state. The baseline policy is $r_0 = 0.30$ and the reform raises the replacement rate to $r_1 = 0.35$.

I report two welfare comparisons. The stationary comparison evaluates each policy at its own invariant distribution, while the fixed-distribution comparison holds the baseline distribution fixed. Neither calculation includes the transition path between stationary equilibria.

The two economies Two economies are compared. **B** is the estimated model and **A** is a one-generation incomplete-markets economy with a single discount factor, no mortality or bequests, and the one-component income process of Section A.3.

Model A is recalibrated so that under $r = 0.30$ it shares B's equilibrium interest rate and capital–labor ratio. Their wealth distributions and HtM shares are left untargeted and therefore differ across economies.

Table A7. Stationary equilibria under the two policies

Model	Policy	R	λ^{eff}	K/Y	Gini	Top 10%	HtM
B	$r = 0.30$	1.054565	0.3148	2.1350	0.7282	0.5385	0.1793
	$r = 0.35$	1.054961	0.3691	2.1296	0.7374	0.5461	0.2113
A	$r = 0.30$	1.054565	0.3148	2.1350	0.6634	0.4738	0.1045
	$r = 0.35$	1.055249	0.3694	2.1256	0.6757	0.4826	0.1165

Notes: λ^{eff} is the unemployment benefit as a fraction of mean employed labor income net of the financing tax.

Stationary and fixed-distribution welfare Stationary welfare under replacement rate r is

$$W^{SS}(r) = \int V^r(x)\, d\Phi^r(x). \tag{A.14}$$

The welfare change from r_0 to r_1 therefore incorporates both changes in household values and changes in the long-run distribution of states.

To isolate the first component, define fixed-distribution welfare as

$$W^{FD}(r) = \int V^{r}(x)\, d\Phi^{r_0}(x), \tag{A.15}$$

so that the stationary welfare change can be written as

$$\Delta W^{SS} = \underbrace{\int [V^{r_1}(x) - V^{r_0}(x)]\, d\Phi^{r_0}(x)}_{\text{value effect}} + \underbrace{\int V^{r_1}(x)\, d[\Phi^{r_1}(x) - \Phi^{r_0}(x)]}_{\text{composition effect}}. \tag{A.16}$$

The first term is the fixed-distribution welfare change in levels; the second captures the additional effect of the reform-induced shift in the stationary distribution. This decomposition applies to welfare levels rather than directly to consumption-equivalent variations.

Consumption-equivalent variation I report welfare changes as consumption-equivalent variations. Let $V^{r_0}(x; g)$ denote the value obtained when baseline consumption is scaled by $1 + g$ at every future date. The CEV is the value of g that equates this scaled baseline welfare with welfare under the reform.

For the stationary comparison,

$$\int V^{r_0}(x; g)\, d\Phi^{r_0}(x) = \int V^{r_1}(x)\, d\Phi^{r_1}(x), \tag{A.17}$$

while for the fixed-distribution comparison,

$$\int V^{r_0}(x; g)\, d\Phi^{r_0}(x) = \int V^{r_1}(x)\, d\Phi^{r_0}(x). \tag{A.18}$$

The fixed-distribution calculation is not transition-path welfare. It evaluates the baseline population under the permanent new-policy environment while holding state weights fixed.